

TUCNLP at SemEval-2026 Task 11: Neuro-Symbolic Content Stripping for Debaised Syllogistic Reasoning

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Abstract

In this paper, we present the solution submitted by TUCNLP at SemEval-2026 Task 11: Disentangling Content and Formal Reasoning in Language Models. The task requires predicting the formal validity of categorical syllogisms while minimizing susceptibility to content-driven biases in English and 11 additional languages. We show that a modestly-sized model (Qwen3-8B) can achieve near-perfect logical reasoning when augmented with a multi-stage neuro-symbolic pipeline: LLM-based content stripping with iterative error correction converts natural language to abstract categorical forms, a classical symbolic parser validates against the twenty-four Aristotelian syllogistic forms, and asymmetric confidence thresholds mediate between symbolic and neural decisions. Across the four subtasks (ST1 to ST4), our system achieves accuracy ranging from 91.1% to 100% and bias-penalized ranking scores ($\mathcal{M}$) from 31.8 to 100.0, with the main bottleneck being overconfident neural predictions that bypass symbolic verification.

1 Introduction

The ability to perform formal deductive reasoning is considered paramount for machine intelligence, yet Large Language Models (LLMs) frequently struggle to separate logical validity from the semantic plausibility of a conclusion. This phenomenon, known as the *content effect* or *belief bias*, mirrors cognitive limitations long observed in human psychology (Evans, 2003; Lampinen et al., 2024). In such scenarios, a model may incorrectly accept a fallacious argument simply because the conclusion aligns with real-world knowledge, or reject a valid deduction because the conclusion is counterintuitive (Eisape et al., 2024; Seals and Shalin, 2024).

SemEval-2026 Task 11 (Valentino et al., 2026) addresses this challenge by requiring systems to disentangle content-driven heuristics from formal

reasoning. The task focuses on categorical syllogisms, i.e. logical structures classically consisting of two premises and a conclusion, where the interplay between truth and validity is systematically manipulated to test for deductive consistency.

We combine a classical syllogistic parser with a LoRA fine-tuned Qwen3-8B (8B parameters), built on the insight that *content stripping*, abstracting away semantic content before reasoning, eliminates content bias by construction when paired with symbolic logic. A multi-stage pipeline handles LLM-based stripping with iterative error correction, symbolic validation, and cascaded neural-symbolic verification with asymmetric confidence thresholds. Compared to zero-shot Qwen3-8B, our system reduces Total Content Effect (TCE), the average accuracy gap between plausibility-congruent and plausibility-incongruent quadrants, from 21.1 to 0.0 percentage points on English and from 31.4 to 1.1 percentage points on multilingual data, while improving accuracy by up to 37 percentage points. Our system achieves a perfect score on ST1 (shared by 11/45 teams) and ranks 5th–7th on the remaining three subtasks. Our code is publicly available.¹

2 Background and Related Work

Syllogistic reasoning has roots in classical logic (Łukasiewicz, 1951) and natural logic traditions (van Benthem, 1986; Sánchez-Valencia, 1991). Recent research focuses on LLMs’ systematic conflation of content plausibility with formal validity. Dasgupta et al. (2022) first demonstrated that LLMs exhibit *content effect*, Lampinen et al. (2024) confirmed that models reason less reliably when semantic content opposes correct inference, Eisape et al. (2024) showed that even CoT-prompted models rarely recognize invalid syllogisms, and Kim et al. (2025) traced these failures to specific rea-

¹<https://github.com/ButasRafael/semeval2026-task11>

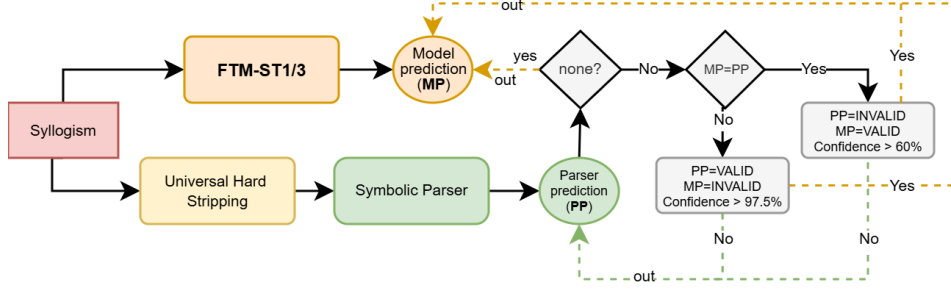


Figure 1: Selective Mode inference (ST1/3). The Symbolic Parser and FT model (FTM) run in parallel producing independent predictions PP and MP, respectively; asymmetric confidence thresholds resolve disagreements.

soning circuits contaminated by world-knowledge attention heads. Benchmarks targeting this problem include SylloBase (Wu et al., 2023) for classical moods, NeuBAROCO (Ozeki et al., 2024) for cross-lingual belief-bias, and the systematic study of Bertolazzi et al. (2024) showing that only supervised fine-tuning reliably mitigates content bias (Wysocka et al., 2025; Seals and Shalin, 2024).

CoT prompting (Wei et al., 2022) and self-consistency decoding (Wang et al., 2023) make intermediate steps explicit but remain susceptible to belief bias (Eisape et al., 2024; Bertolazzi et al., 2024). Quasi-symbolic abstractions (Ranaldi et al., 2025) and fully symbolic CoT methods (Xu et al., 2024; Lyu et al., 2023) translate to formal logic at the cost of formalization overhead. Hybrid neural-symbolic methods include Logic-LM (Pan et al., 2023), which delegates deduction to external solvers, and PEIRCE (Quan et al., 2025), which unifies material and formal inference through iterative refinement. Our system builds on these foundations by using LLM-based content stripping to bridge the formalization gap, converting natural language syllogisms into abstract categorical forms for classical validation, with a fine-tuned neural fallback for unresolvable cases.

3 System Overview

Our system is a hybrid neuro-symbolic architecture that combines a classical syllogistic parser with a LoRA (Hu et al., 2022) fine-tuned Qwen3-8B (Qwen Team, 2025). The central design insight is that *content stripping*, i.e., abstracting away semantic content before reasoning, directly diminishes the content effect. The symbolic component, a classical syllogistic parser backed by Z3 SMT (De Moura and Bjørner, 2008) verification, reasons over abstract logical form with zero content bias, while the neural component handles cases the

parser cannot resolve.

3.1 Data Preparation

Content Stripping. We apply LLM-based content stripping using zero-shot Qwen3-8B with in-context examples (8 demonstrations for hard stripping; see Appendix B) in two modes: *soft stripping* replaces content terms with variables A , B , C while preserving grammar, and *hard stripping* normalizes each sentence to one of four categorical forms (*All/No/Some/Some...not X are Y*). A regex based post-processing pipeline handles set-theoretic expressions, double negation, modals, and quantifier variants. When stripping fails (i.e., the output does not parse into exactly three well-formed categorical sentences), up to 3 retries with targeted error feedback are attempted: the system diagnoses the specific failure mode (incorrect sentence count, malformed variables, missing conclusion marker, compound predicates, or invalid syllogistic figure) and generates corrective hints for each retry, with a thinking-mode fallback on the final attempt.

Variant Generation and Augmentation. Each syllogism is expanded into three variants (original, soft-stripped, hard-stripped) for symbolic verification and training augmentation. For ST2/4, we add 3–5 LLM-generated filler premises (semantically related, logically irrelevant) in indexed format $[0] \dots [N]$ to train premise selection. For ST3/4, we translate into 11 target languages via DeepL² and Google Translate³, preserving quantifiers and conclusion markers.

3.2 Fine-Tuning

We fine-tune Qwen3-8B using LoRA (Hu et al., 2022) with DoRA (Liu et al., 2024) applied to all at-

²<https://www.deepl.com>

³<https://translate.google.com>

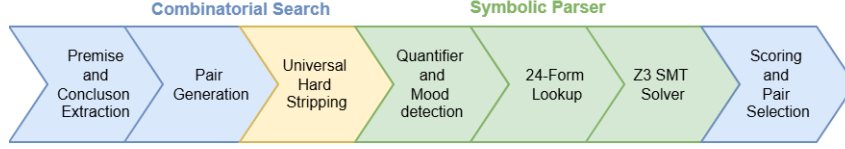


Figure 2: Combinatorial premise search and symbolic parsing pipeline (ST2/4). Premise pairs are hard-stripped, validated against the 24 syllogistic forms (with Z3 fallback), and ranked by a multi-factor scoring function.

tention and FFN projection modules. We train four specialized adapters, one per subtask (FTM-ST1 to FTM-ST4): validity-only adapters for ST1 and ST3 use a lower LoRA rank, while the joint validity and premise retrieval adapters for ST2 and ST4 use a higher rank with RSLoRA (Kalajdzievski, 2023) to accommodate the increased task complexity. A key training design is **quadrant-balanced sampling**: each batch contains equal representation from all four validity $\times$ plausibility quadrants (VP, VI, IP, II), directly minimizing TCE during training. For the multilingual subtasks (ST3/4), we extend this with joint quadrant-language stratification. All hyperparameter values are provided in Appendix A.

3.3 Inference: Selective Mode (ST 1 & 3)

For binary validity classification, we use a **parser-primary** architecture with asymmetric neural fallback (Figure 1). Each input is processed along two parallel paths:

(1) Symbolic path. The syllogism is stripped to hard categorical form (for ST3, a *universal stripper* directly converts any language to English categorical form via multilingual LLM prompting) and is then passed to the Symbolic Parser. For cases where the standard lookup fails, a Z3 SMT solver (De Moura and Bjørner, 2008) provides first-order logic verification with existential import (implied in Aristotelian logic, which the task at hand is based on).

(2) Neural path. The fine-tuned model (FTM) predicts validity and extracts a *token-level softmax confidence*, i.e. the softmax probability at the generated validity token position, summed over all tokenizer variants of each label.

Decision logic. If path (1) returns None (parse failure), we fall back to path (2). If both agree, we use the parser result. On disagreement, we apply *asymmetric confidence thresholds*, calibrated on the development split: since the FTM is only 45% accurate when contradicting a parser VALID judgment, override to INVALID requires $\text{conf} \geq 0.975$; conversely, the FTM is 100% accurate when con-

tradicting parser INVALID, so override to VALID requires only $\text{conf} \geq 0.60$. Below these thresholds, the parser decision stands.

3.4 Inference: Cascaded Verifier (ST 2 & 4)

For joint validity and premise retrieval, we use a three-stage **cascaded verifier** (Figure 3). Because distractor premises share the domain of true premises, content stripping can collapse near-synonymous terms onto the same variable, producing spurious symbolic matches; we therefore adopt an FT-primary design where the symbolic pipeline primarily identifies correct premises rather than classifying validity:

Stage 1: FTM prediction. The fine-tuned model outputs validity, relevant premise indices, and confidence. **Stage 2: Symbolic verification.** If the model predicts INVALID with high confidence (threshold in Appendix A), we return immediately. Otherwise, combinatorial symbolic search generates all $\binom{N}{2}$ premise pairs from the N available premises (Figure 2). A *connectivity gate* pre-filters pairs lacking shared terms via lemmatized overlap (relaxed for non-Latin scripts where lemmatization is unreliable). Each pair is stripped to hard categorical form and checked against the 24 valid forms (with Z3 fallback). When multiple valid pairs are found, a scoring function ranks them by method quality (deterministic match vs. Z3), figure determinacy, FT hint agreement, and for ST4, conclusion alignment via BGE-M3 (Chen et al., 2024).

For ST2, if a valid pair is found, we override to VALID with the symbolic premises. For ST4, a *semantic guard* using BGE-M3 hybrid scoring (dense + ColBERT) must additionally confirm sufficient conclusion term coverage before accepting the override.

Stage 3: Valid premise verification. If the model predicts VALID, we verify the selected premises via symbolic search. For ST4, per-language confidence thresholds select whichever premise pair achieves better conclusion term coverage. A *double-negative safety check* overrides to

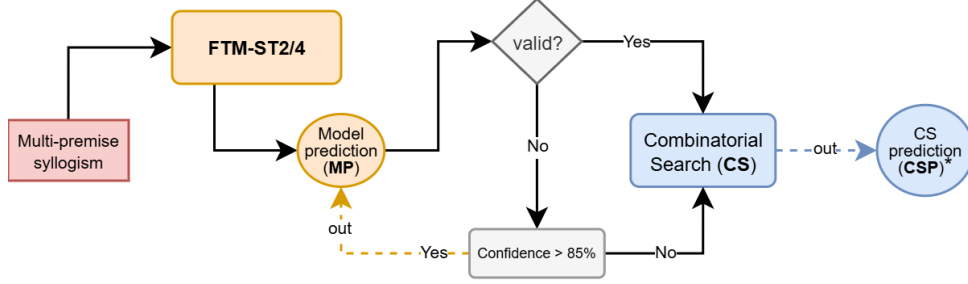


Figure 3: Cascaded Verifier inference (ST2/4). MP = FTM prediction; CSP = combinatorial search prediction. *ST4 adds semantic and per-language guards.

INVALID if both premises are negative.

4 Experimental Setup

Data. The organizers provide 960 English syllogisms balanced across four validity $\times$ plausibility quadrants (Valentino et al., 2026). Training data is English-only; multilingual test data covers 11 additional languages. We apply an 85/15 stratified train/validation split for fine-tuning, preserving quadrant balance. After the augmentation described in §1, ST1 trains on the original English split (809 examples); ST2 additionally incorporates distractor premises (811 training examples); ST3/4 expands to 12 languages (9,792 for ST3; 9,732 for ST4). The test sets contain 191 (ST1), 190 (ST2), and 192 instances each (ST3/4, approximately 17–18 per language, in 11 target languages). Fine-tuning hyperparameters are listed in Appendix Table 3.

Evaluation. Following the official protocol (Valentino et al., 2026), all subtasks are ranked by $\mathcal{M} = S/(1 + \ln(1 + \text{TCE}))$, where S is accuracy for ST1/3 and $\text{Avg}(\text{ACC}, \text{F1}_{\text{premises}})$ for ST2/4. For ST3/4, TCE additionally includes a cross-lingual stability penalty. To isolate the contribution of our pipeline, we compare against the same Qwen3-8B model in a zero-shot setting: identical prompt format but without fine-tuning, content stripping, or symbolic verification.

5 Results

5.1 Main Results

Table 1 presents the performance obtained on the official test set. Our full system achieves perfect accuracy (100.0%), with zero content effect on ST1, and near-perfect results on ST3 (98.4% on the 11-language test set). Compared to the same model in zero-shot mode, the ranking metric $\mathcal{M}$ improves by factors of $5.2\times$ (ST1), $3.2\times$ (ST2), $4.2\times$ (ST3),

Subtask	System	ACC	F1 _p	TCE↓	$\mathcal{M}$
ST1	Zero-shot	78.0	–	21.1	19.0
	Ours	100.0	–	0.0	100.0
ST2	Zero-shot	78.6	56.3	18.8	16.9
	Ours	96.9	95.8	1.2	54.2
ST3	Zero-shot	60.9	–	31.4	13.6
	Ours	98.4	–	1.1	56.7
ST4	Zero-shot	76.0	52.3	11.6	18.2
	Ours	91.1	90.1	5.3	31.8

Table 1: Official test results. Zero-shot: Qwen3-8B without fine-tuning, stripping, or symbolic verification. F1_p: premise retrieval F1. $\mathcal{M}$: official ranking metric (§4). Best in **bold**.

and $1.8\times$ (ST4). On the official leaderboard, the system achieves the maximum possible score on ST1 (shared by 11 of 45 teams) and ranks 5th/16 on ST2, 6th/15 on ST3, and 7th/15 on ST4.

The results reveal a clear difficulty gradient: English validity-only (ST1) is fully solved; adding either multilinguality (ST3) or premise retrieval (ST2) introduces small residual errors; combining both (ST4) yields the most errors. Across all subtasks, TCE drops dramatically, from 21.1 to 0.0 (ST1) and from 31.4 to 1.1 (ST3), confirming that content stripping and symbolic verification effectively eliminate content bias. A component-level ablation (Appendix F) shows that fine-tuning alone improves accuracy but retains moderate TCE, while the Symbolic Parser provides near-zero TCE; the hybrid pipeline combines both advantages.

5.2 Content Effect Analysis

Table 2 breaks down the accuracy by the validity $\times$ plausibility quadrant. The zero-shot model exhibits the classic content effect pattern: accuracy is highest on Invalid+Implausible instances (where content and logic align in rejecting the argument) and lowest on Valid+Implausible (where the

Quadrant	ST1 (EN)		ST3 (ML)	
	ZS	Full	ZS	Full
Valid + Plaus.	81.2	100.0	50.0	97.9
Valid + Implaus.	60.4	100.0	29.2	97.9
Invalid + Plaus.	74.5	100.0	71.7	97.8
Invalid + Implaus.	95.8	100.0	92.0	100.0
TCE	21.1	0.0	31.4	1.1

Table 2: Per-quadrant accuracy (%) for zero-shot (ZS) vs. our full system on ST1 and ST3.

conclusion is counter-intuitive but logically valid). On ST1, the zero-shot gap between these two quadrants is 35.4 percentage points (95.8% vs. 60.4%). Our full system eliminates this asymmetry entirely: all four quadrants reach 100.0%.

On the multilingual ST3, the zero-shot content effect is even more pronounced (92.0% vs. 29.2%, a 62.8-point gap). Our system reduces this to a maximum 2.2-point gap, with the small residual driven by 3 errors in Swahili (2) and Russian (1). For ST4, the residual TCE of 5.3 is partly inflated by annotation inconsistencies (§5.3) and partly reflects the added difficulty of premise selection from distractor-rich inputs. Per-language results are in Appendix C.

5.3 Error Analysis

Across all four subtasks, the system produces 24 apparent validity errors and 9 premise-selection errors against the official ground truth (breakdown in Appendix D). On ST1, all 191 instances are classified correctly, with the Symbolic Parser and fine-tuned model agreeing on every prediction (only 2 cases require Z3 fallback). However, manual verification of each error against the 24 classically valid Aristotelian forms (assuming existential import) suggests that **12 of the 24 apparent validity errors reflect annotation inconsistencies** in the test set rather than system failures, leaving only 12 genuine system errors.

Of the 12 identified inconsistencies (detailed in Appendix G), 10 are syllogisms labeled invalid that instantiate classically valid Aristotelian forms (7 unconditionally valid, 3 requiring existential import), and 2 are labeled valid but are in fact invalid. These concentrate in ST4 (10 of 12), reflecting difficulties pertaining to multilingual annotation.

Among the 12 genuine validity errors, the dominant type is **high-confidence FTM false negatives** (7/12 = 58%), where the fine-tuned model predicts invalid with confidence above the bypass thresh-

old, short-circuiting symbolic verification entirely. These span multiple languages and both plausibility conditions, suggesting content-independent reasoning failures rather than bias-driven mistakes. The remaining 5 genuine errors are false positives: 3 from incorrect content stripping producing a spurious valid form and 2 from FTM overrides where the neural prediction overruled a correct symbolic rejection. After accounting for annotation inconsistencies, no single language dominates the ST4 error distribution, and content stripping accounts for only 25% of genuine errors.

6 Conclusion

We introduce a hybrid neuro-symbolic system for SemEval-2026 Task 11 that combines a LoRA fine-tuned Qwen3-8B with a classical syllogistic parser operating on content-stripped logical forms. By decomposing inference into a multi-stage pipeline with iterative error correction, asymmetric confidence thresholds, and classical logic safety checks, the system achieves perfect accuracy on English ST1 (shared by 11/45 teams) and ranks 5th–7th on ST2–ST4. Our results demonstrate that a modestly-sized 8B model, augmented with symbolic reasoning over content-stripped forms, can largely eliminate the content bias that persists even in much larger models under zero-shot prompting. Among genuine errors, the primary bottleneck is overconfident neural predictions (58%) that bypass symbolic verification; improving FTM calibration and applying activation steering (Valentino et al., 2025; Maraia et al., 2026) are the most promising directions for further gains.

Limitations

Our approach has several limitations. The Symbolic Parser handles only classical categorical syllogisms (two premises, three terms, four quantifier types) and cannot extend to multi-step reasoning or non-standard quantifiers. Content stripping depends on LLM-based formalization, which is difficult to debug for low-resource and morphologically complex languages (Swahili, Bengali, Telugu). The test sets contain only 17 to 18 instances per language, making per-language estimates statistically noisy. Finally, our identification of 12 annotation inconsistencies (§5.3) assumes Aristotelian logic with existential import; under modern Boolean semantics, 3 of these (Darapti and subalternation) would be reclassified as genuine errors.

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A Implementation Details

Table 3 lists the per-subtask hyperparameters for all four adapters.

Hyperparameter	FM1	FM2	FM3	FM4
LoRA rank r	16	32	16	32
LoRA α	32	64	32	64
RSLoRA	–	Yes	–	Yes
Epochs	4	4	2	2
Batch / GPU	4	4	4	4
Grad. accum.	8	8	8	8
Eff. batch size	32	32	32	32
Learning rate	1e-4	1e-4	5e-5	5e-5
Max seq. length	1024	1024	1024	1536
Training examples	809	811	9,792	9,732
GPUs (V100-32GB)	2	4	2	4

Table 3: Per-subtask hyperparameters. All adapters use Qwen3-8B with DoRA, LoRA dropout 0.1, AdamW optimizer, cosine LR schedule with 10% warmup, weight decay 0.1, and FP16 mixed precision. Early stopping monitors the primary metric.

Training. All adapters are trained with AdamW, cosine learning rate schedule with 10% warmup, and FP16 mixed precision on 2 to 4 NVIDIA Tesla V100-SXM2-32GB GPUs using PyTorch DDP.⁴ English subtasks (ST1/2) train for 4 epochs with learning rate 10^{-4} ; multilingual subtasks (ST3/4) train for 2 epochs at 5×10^{-5} over the larger translated corpus. Early stopping monitors the official ranking metric on the held-out evaluation split. Training ranges from approximately 2 hours (ST1) to 6 hours (ST4).

Inference thresholds. All confidence thresholds are calibrated on the development split. For the cascaded verifier (ST2/4): the high-confidence INVALID bypass threshold is 0.85; the BGE-M3 semantic guard requires ≥ 0.93 relative conclusion term coverage; and per-language FT confidence thresholds range from 0.80 (Bengali, Telugu) to 0.95 (high-resource Latin-script languages), with intermediate values for Russian (0.93), Chinese (0.85), and Swahili (0.88).

B Data Preparation Examples

Table 4 illustrates the two content stripping modes on syllogisms from opposite content effect quadrants. The first example is *invalid but plausible*: its conclusion aligns with common sense, tempting a model to predict valid. The second is *valid but implausible*: its absurd content tempts a model to reject a logically sound argument. In both cases, hard stripping reduces the input to abstract categorical form, enabling the symbolic parser to determine validity without content interference.

⁴PyTorch 2.5, HuggingFace Transformers 4.57, PEFT 0.18, Accelerate 1.12, Z3 4.15, and BGE-M3 embeddings.

Example 1: Invalid + Plausible (content $\rightarrow$ false positive)	
Original	All cars are a type of vehicle. No animal is a car. Therefore, no animal can be a vehicle.
Soft	All A are a type of B. No C is an A. Therefore, no C can be a B.
Hard	All A are B. No C are A. Therefore, No C are B.
Example 2: Valid + Implausible (content $\rightarrow$ false negative)	
Original	Every cat is an invisible creature. A number of cats are animals. Consequently, a portion of animals are invisible.
Soft	Every A is a B. A number of A are C. Consequently, a portion of C are B.
Hard	All A are B. Some A are C. Therefore, Some C are B.

Table 4: Content stripping examples. Soft stripping preserves grammar while replacing content terms with variables; hard stripping normalizes to canonical categorical form.

Table 5 shows two examples of LLM-augmented premise sets used for ST2/4 training. Each syllogism is expanded with 4–5 thematically related but logically irrelevant filler premises (shown in gray). The model must identify the relevant premise pair (bold) and determine validity.

Example 1: Valid, relevant premises = {4, 5}	
[0]	There exist some vehicles that are neither cars nor motorcycles.
[1]	A portion of vehicles are electric.
[2]	Some vehicles are used for long-distance travel.
[3]	Every vehicle is either a car, a motorcycle, or something else entirely.
[4]	Nothing that is a car is a motorcycle.
[5]	A few things called vehicles are identified as being cars.
[C]	Therefore, some vehicles are not motorcycles.
Example 2: Invalid, relevant premises = {}	
[0]	Every person in the city has at least one profession.
[1]	A portion of professionals work more than one job.
[2]	No doctor is a lawyer.
[3]	Lawyers are, without exception, professionals.
[4]	There exist people who are both engineers and artists.
[5]	Some teachers are also part-time counselors.
[6]	All doctors have a medical degree.
[C]	Therefore, a number of professionals are doctors.

Table 5: LLM-augmented premise sets for ST2/4. Gray: filler premises. Bold: relevant premises (empty set for invalid syllogisms). In Example 2, premises [2] and [3] are the original premises but yield an invalid conclusion (undistributed middle).

Hard Stripping Prompt Structure. The hard stripping prompt instructs the LLM to convert each syllogism into exactly three categorical sentences using variables A , B , C . It specifies the four valid output forms (*All/No/Some/Some...not X are*

Y), provides 8 in-context demonstrations covering all four quantifier types, and includes detailed negation-handling rules: (1) *not all/every* $\rightarrow$ *Some X are not Y*; (2) *not any* $\rightarrow$ *No X are Y*; (3) double negatives cancel (*no X are not Y* $\rightarrow$ *All X are Y*); (4) predicate negation (*every X is not Y* $\rightarrow$ *No X are Y*).

Error Diagnosis on Retry. When stripping fails validation, the system classifies the failure and generates targeted corrective hints before each retry:

- *Sentence count* $\neq$ 3: split or merged premises.
- *Malformed variables*: variables beyond $A/B/C$, or variables appearing only in the conclusion.
- *Missing conclusion*: output lacks a “Therefore,” marker.
- *Compound predicates*: e.g., “All A are both B and C.”
- *Invalid figure*: no shared middle term across premises.

C Additional Results

Table 6 shows the per-language validity accuracy for the multilingual subtasks. On ST3, our system achieves 100% on 9 of 11 languages; only Swahili (88.2%) and Russian (94.1%) have errors. On ST4, Portuguese is the only error-free language. The worst-performing language against the official ground truth is Russian (76.5% on ST4, 4 of 17 apparent errors), followed by Swahili (82.4%). However, as discussed in §5.3, manual verification suggests that 3 of Russian’s 4 ST4 errors are annotation inconsistencies (the syllogisms instantiate valid Barbara or Darapti forms), leaving only 1 genuine system error for Russian. After accounting for these inconsistencies, the per-language error distribution in ST4 is more uniform, with no single language accounting for more than 2 genuine errors.

Lang	ST3		ST4	
	ZS	Full	ZS	Full
it	61.1	100.0	66.7	88.9
es	55.6	100.0	77.8	94.4
fr	66.7	100.0	77.8	94.4
zh	61.1	100.0	50.0	94.4
de	66.7	100.0	83.3	94.4
te	82.4	100.0	82.4	94.1
sw	47.1	88.2	88.2	82.4
pt	52.9	100.0	88.2	100.0
nl	64.7	100.0	94.1	94.1
bn	52.9	100.0	58.8	88.2
ru	58.8	94.1	70.6	76.5

Table 6: Per-language validity accuracy (%) on ST3 and ST4. ZS = zero-shot Qwen3-8B; Full = our full system. Each language has 17 to 18 test instances.

D Error Breakdown

Table 7 categorizes the 33 errors (24 validity + 9 premise-only) across subtasks. ST1 has zero errors and is omitted.

Error Type	ST2	ST3	ST4	Total
GT annotation error	2	0	10	12
Strip-induced FP	0	1	2	3
FT-override FP	0	0	2	2
High-conf. FT FN	2	2	3	7
Genuine validity errors	2	3	7	12
Premise-only	2	–	7	9

Table 7: Error categorization across subtasks (ST1 has 0 errors). Top row: annotation inconsistencies where our predictions match valid Aristotelian forms.

E Out-of-Domain Evaluation: NeuBAROCO

To assess generalization beyond the shared task data, we evaluate our Selective Mode pipeline (ST1 adapter with Qwen3-8B) on NeuBAROCO (Ozeki et al., 2024), an independent benchmark of 899 English syllogisms designed to probe belief-bias effects. We compare against three frontier models in zero-shot mode (no fine-tuning, no symbolic verification): GPT-5, Claude, and Gemini (with and without thinking mode). All models receive identical prompting.

System	ACC	TCE↓	$\mathcal{M}$
GPT-5 (zero-shot)	84.97	12.55	23.56
Claude (zero-shot)	89.21	12.80	24.61
Gemini (zero-shot)	92.44	4.13	35.07
Gemini + thinking	95.00	6.29	31.81
Ours (Qwen3-8B + pipeline)	91.77	5.27	32.36

Table 8: Out-of-domain evaluation on NeuBAROCO (899 instances). Zero-shot models use identical prompting without fine-tuning or symbolic verification. $\mathcal{M} = \text{ACC}/(1 + \ln(1 + \text{TCE}))$. Our system achieves the highest raw accuracy among all models except Gemini variants, while substantially reducing content effect compared to Claude and GPT. These tests were done with an initial variant of our system so it’s fair to assume that the latest version would obtain even better results

Although Gemini with thinking achieves higher raw accuracy (95.0% vs. 91.8%), it retains substantial content bias (TCE = 6.29), which reduces its combined score to 31.8. Our system achieves the best combined score (32.4) by pairing high accuracy with a significantly lower content effect (TCE = 5.27) than any zero-shot baseline, confirming that content stripping and symbolic verification generalize beyond the training distribution.

F Ablation Study

Table 9 isolates the contribution of each pipeline component by evaluating partial configurations against the full system: (1) *Zero-shot*: Qwen3-8B with no fine-tuning or symbolic verification; (2) *FT-only*: LoRA-adapted model without the symbolic parser (ST1–3; no pure FT-only run exists for ST4); (3) *Symbolic-only*: symbolic pipeline without fine-tuning (ST1–3; ST2 uses Qwen3-32b for content stripping); (4) *FT-combin.*: FT model performing combinatorial premise-pair search without symbolic verification (ST4 only).

The results reveal clear complementarity between the neural and symbolic components. Fine-tuning alone dramatically improves accuracy over zero-shot (from 60.9–78.6% to 96.8–99.0% on ST1–3) and premise retrieval F1 (from 56.8% to 86.6% on ST2), but retains moderate content effects (TCE 2.1–3.1). Notably, the symbolic-only pipeline achieves higher premise F1 than FT-only on ST2 (89.5% vs. 86.6%), suggesting that combinatorial symbolic search is more precise at identifying relevant premises than direct neural prediction. However, it cannot handle all inputs, particularly in multilingual settings (7 parse failures on ST3,

Config	ST1		ST2			ST3		ST4		
	ACC	TCE↓	ACC	F1 _p	TCE↓	ACC	TCE↓	ACC	F1 _p	TCE↓
Zero-shot	78.0	21.1	78.6	56.8	18.8	60.9	31.4	76.0	52.3	11.6
FT-only	99.0	2.1	96.8	86.6	3.1	97.9	2.2	—	—	—
Symbolic-only	97.9	2.1	92.6 [‡]	89.5 [‡]	8.0 [‡]	96.2 [‡]	2.5 [‡]	—	—	—
FT-combin. [§]	—	—	—	—	—	—	—	87.0	82.8	11.7
Full pipeline	100.0	0.0	96.9	95.8	1.2	98.4	1.1	91.1	90.1	5.3

Table 9: Component ablation: validity accuracy (%), premise retrieval F1 (F1_p), and TCE across pipeline configurations. [†]Computed over 185/192 parseable items; 7 unparsed are all non-English. [‡]Uses Qwen3-32b for content stripping. [§]FT-combin.: FT model evaluates all $\binom{N}{2}$ premise pairs without the symbolic parser; no pure FT-only or symbolic-only run exists for ST4.

all non-English). For ST4, an FT-combinatorial baseline (FT model evaluating all premise pairs without symbolic verification) achieves 87.0% accuracy but with TCE (11.7) comparable to the zero-shot baseline (11.6), indicating that content bias persists without symbolic verification. The full pipeline achieves the best of both: accuracy and F1 meet or exceed any single component, while TCE drops to 0.0–5.3. The gain is most dramatic on ST1, where $\mathcal{M}$ rises from 46.3 (FT-only) to 100.0 (full pipeline), a $2.2\times$ improvement driven entirely by TCE elimination. It is worth noting that, due to the relatively small number of test samples, some score differences may appear modest, but they are nonetheless consistent across all test setups.

G Annotation Inconsistency Details

Table 10 lists the 12 test instances identified as annotation inconsistencies in §5.3. For each, we show the ground-truth label, the Aristotelian form instantiated by the relevant premises and conclusion (abstracted to variables A, B, C, D), and the subtask/language. Forms marked with [†] require existential import, standard in Aristotelian logic. $\nRightarrow$ denotes invalid inference.

False negatives. The 10 entries below show the full English surface form of valid syllogisms mislabeled as invalid. **Major premise**, **minor premise**, and **conclusion** are highlighted; filler premises are in black. [†]Valid only under existential import (Aristotelian logic).

1. a660d443 (ST4/ru, Barbara AAA-1)

Any snail moves with great speed. Every single sloth is a fast runner. It is true that all starfish are fast. **Anything that is an animal is fast.** Some clams are swift animals. **Every single turtle is an animal.** $\therefore$ **There are no turtles that are not fast.**

2. 9d57ec74 (ST4/de, Celarent EAE-1)

All zoos are empty of life. No mammal has a heart. **Not a single creature that is a dog is an animal.** Some bears are actually robots. **Every single mammal is a dog.** $\therefore$ **No mammal is an animal.**

3. 827501ed (ST4/es, Barbara AAA-1)

Every single broccoli is a vegetable. It is true that all fruits are edible. Any potato is a vegetable. Some edible things are grains. There are a few roots that are edible. **Every single vegetable is edible.** **Anything that is a carrot is a vegetable.** $\therefore$ **There are no carrots that are not edible.**

4. 475136be (ST4/it, Barbara AAA-1)

Every single elephant is capable of flight. There are no hippos that cannot fly. Any creature that is a dog can fly. **Anything which is an animal is capable of flight.** **Every single insect is an animal.** $\therefore$ **There are no insects that cannot fly.**

5. 25d671a3 (ST4/ru, Baroco AOO-2)

Every single person has three heads and five arms. It is the case that books eat the students who read them. No human has ever learned to read or write. **Every person is a human.** Some students are actually alien robots. **At least one student is not a human.** $\therefore$ **There is at least one student is not a person.**

6. 4c62aa84 (ST4/sw, Baroco AOO-2)

Many animals are kept by humans as companions. Some cats prefer to sleep for many hours a day. Every veterinarian treats various types of pets. **Every single dog has fur.** It is true that dogs are often called man’s best friend. **At least one cat does not have fur.** $\therefore$ **At least one cat is not a dog.**

ID	ST / Lang	GT	Form	Premises $\Rightarrow$ Conclusion
<i>Labeled invalid, instantiate valid forms (10)</i>				
a660d443	ST4 / ru	inv	Barbara	All A are B; All C are A $\Rightarrow$ All C are B
9d57ec74	ST4 / de	inv	Celarent	No A is B; All C are A $\Rightarrow$ No C is B
827501ed	ST4 / es	inv	Barbara	All A are B; All B are C $\Rightarrow$ All A are C
475136be	ST4 / it	inv	Barbara	All A are B; All C are A $\Rightarrow$ All C are B
25d671a3	ST4 / ru	inv	Baroco	All A are B; Some C are not B $\Rightarrow$ Some C are not A
4c62aa84	ST4 / sw	inv	Baroco	All A are B; Some C are not B $\Rightarrow$ Some C are not A
a3a4fcb5	ST4 / it	inv	Celarent	No A is B; All C are A $\Rightarrow$ No C is B
37190acf	ST2 / en	inv	Subaltern [†]	All A are B $\Rightarrow$ Some B are A
612f04e8	ST2 / en	inv	Subaltern [†]	All A are B $\Rightarrow$ Some A are B
0a7f2135	ST4 / ru	inv	Darapti [†]	All A are B; All A are C $\Rightarrow$ Some C are B
<i>Labeled valid, actually invalid (2)</i>				
c6c1a6bc	ST4 / nl	val	Undist. mid.	All A are B; All C are B $\nRightarrow$ All C are A
66efe831	ST4 / sw	val	4th-term	No A is B; Some B are C $\nRightarrow$ Some A are not D

Table 10: The 12 annotation inconsistencies identified in the test set (§5.3). [†]Valid only under existential import (Aristotelian logic). The first 10 are false negatives (valid syllogisms mislabeled as invalid); the last 2 are false positives.

7. **a3a4fcb5** (ST4/it, Celarent EAE-1)
 Not a single thing that is a vertebrate is a mammal. Some animals lay eggs in jelly. Every ecosystem has a food chain. It is true that skin can be permeable. Every single frog is a vertebrate. $\therefore$ It must be the case that no frog is a mammal.
8. **37190acf** (ST2/en, Subaltern[†])
 Every pea is a vegetable. Some beans are vegetables. There are many beets that are vegetables. Any radish is a vegetable. It is a fact that all turnips are vegetables. It is a fact that every potato is a vegetable. It is also true that all carrots are vegetables. $\therefore$ A few vegetables are carrots.
9. **612f04e8** (ST2/en, Subaltern[†])
 Some hinges are bones of a skeleton. All of the things that are doors are planets. There are no frames that are not made of flesh. It is true that all knobs are eyes of a beast. Any house that is built is actually a zoo. All of the things that are windows are planets. $\therefore$ It can be deduced that there are windows that are planets.
10. **0a7f2135** (ST4/ru, Darapti[†])
 Some turtles are creatures capable of flight. Every single lizard is a mammal. It is the case that all frogs are reptiles. There are no

reptiles that are cold-blooded. Anything that is a snake is a bird. It is a fact that all snakes are reptiles. $\therefore$ It follows that a few reptiles are birds.

False positives. The 2 entries below show valid-labeled syllogisms that are actually invalid. Conclusions are highlighted; premises are in black to emphasize that no valid inference leads to the conclusion.

1. **c6c1a6bc** (ST4/nl, Undistributed Middle)
 All humans are mammals with backbones. Any creature that is a dolphin is a mammal. Anything that has a backbone is a mammal. There are no dogs that do not have a backbone. It is true that some animals with backbones are reptiles. Every single fish has a backbone. Every single whale is a mammal. $\therefore$ Every whale has a backbone.
2. **66efe831** (ST4/sw, Fourth Term)
 It is a fact that animals adapt. Every zoo cares for wildlife. All farms have livestock. Most pets live indoors with humans. It is not true that any aquatic animals are cats. A portion of cats are domesticated animals. $\therefore$ There are some aquatic animals that are not wild animals.